

Data Structures



Lecture 19 Tree Problems

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Tree Problems

Check is Two Binary Trees are **Identical**

Tree Problems

```
def identicalTrees(a, b):  
  
    # 1. Both empty  
    if a is None and b is None:  
        return True  
  
    # 2. Both non-empty -> Compare them  
    if a is not None and b is not None:  
        return ((a.data == b.data) and  
                identicalTrees(a.left, b.left) and  
                identicalTrees(a.right, b.right))  
  
    # 3. one empty, one not -- false  
    return False
```

Tree Problems

Check if a Binary Tree is **Balanced**

Tree Problems

```
def isBalanced(root):  
  
    # Base condition  
    if root is None:  
        return True  
  
    # for left and right subtree height  
    lh = height(root.left)  
    rh = height(root.right)  
  
    # allowed values for (lh - rh) are 1, -1, 0  
    if (abs(lh - rh) <= 1) and isBalanced(  
        root.left) is True and isBalanced(root.right) is True:  
        return True  
  
    # if we reach here means tree is not  
    # height-balanced tree  
    return False
```

Tree Problems

Check if a Binary Tree has Duplicate Elements

Tree Problems

```
# To check if tree has duplicates
def checkDup( root) :

    s=set()
    return checkDupUtil(root, s)
```

Tree Problems

```
def checkDupUtil( root, s) :  
  
    # If tree is empty, there are no  
    # duplicates.  
    if (root == None) :  
        return False  
  
    # If current node's data is already present.  
    if root.data in s:  
        return True  
  
    # Insert current node  
    s.add(root.data)  
  
    # Recursively check in left and right  
    # subtrees.  
    return checkDupUtil(root.left, s) or checkDupUtil(root.right, s)
```


Tree Problems

Find the **Unique Elements** in a Binary Tree

Tree Problems

Find the **Duplicate Elements** in a Binary Tree

Tree Problems

Find Occurance of an Element in a Binary Tree

Tree Problems

Find Occurance of All Elements in a Binary Tree

Tree Problems

Check if a Binary Tree is Skewed

each node has only one child node or none.

Tree Problems

```
def isSkewedBT(root):  
  
    # check if node is None or is a leaf node  
    if (root == None or (root.left == None and  
                          root.right == None)):  
        return 1  
  
    # check if node has two children if  
    # yes, return false  
    if (root.left and root.right):  
        return 0  
    if (root.left) :  
        return isSkewedBT(root.left)  
    return isSkewedBT(root.right)
```

Tree Problems

Print all the **Full Nodes** of a Binary Tree

both left and right children are non-empty.

Tree Problems

Print all the Nodes with 1 Child of a Binary Tree

Tree Problems

Build a Balanced BST from a Sorted Array

Tree Problems

```
def sortedArrayToBST(arr):  
  
    if not arr:  
        return None  
  
    # find middle index  
    mid = (len(arr)) // 2  
  
    # make the middle element the root  
    root = Node(arr[mid])  
  
    # left subtree of root has all  
    # values < arr[mid]  
    root.left = sortedArrayToBST(arr[:mid])  
  
    # right subtree of root has all  
    # values > arr[mid]  
    root.right = sortedArrayToBST(arr[mid+1:])  
    return root
```

Tree Problems

Given a BST Print Elements in Descending Order

Do yourself

Tree Problems

Given a BST find Kth Largest Element

Tree Problems

```
def kthLargestUtil(root, k, c):  
    if root == None or c[0] >= k:  
        return  
  
    # Follow reverse inorder traversal  
    # so that the largest element is  
    # visited first  
    kthLargestUtil(root.right, k, c)  
  
    # Increment count of visited nodes  
    c[0] += 1  
  
    # If c becomes k now, then this is  
    # the k'th largest  
    if c[0] == k:  
        print("K'th largest element is",  
              root.key)  
        return  
  
    # Recur for left subtree  
    kthLargestUtil(root.left, k, c)
```

Tree Problems

```
# Function to find k'th largest element
def kthLargest(root, k):

    # Initialize count of nodes
    # visited as 0
    c = [0]

    # Note that c is passed by reference
    kthLargestUtil(root, k, c)
```